

Analytic characterization of NIHAO dark-matter halo profiles

C. Accard A. Chartier A. Ruch

DU MARI

27th April 2026

Cosmological framework and motivation

- In the Λ CDM paradigm, dark matter haloes are the gravitational scaffolding within which baryons cool and galaxies form.
- Their internal structure affects galaxy morphology, star formation history, kinematics, and feedback response.
- High resolution cosmological simulations such as NIHAO provide detailed halo density profiles across a broad mass range.
- The goal here is not only to fit these profiles, but to identify analytic forms that remain useful for theory and observations.

Main scientific issue

How do we move from numerical halo profiles to compact laws that are both **accurate** and **analytically tractable**?

Why analytic halo profiles are needed

- Many applications require evaluation of derived quantities rather than the density itself.
- They also allow asymptotic analysis, parameter inference, and easier physical interpretation.
- Purely numerical tabulations are often too costly or too opaque for these purposes.

Key point

A good halo law should not only fit $\rho(r)$ well; it should also keep $M(< r)$, v_{circ} , Φ , and projected quantities usable.

Automatic analytic characterization

- Manual profile fitting becomes inefficient, subjective, and difficult to reproduce as samples grow.
- Automatic characterization provides a systematic search over functional forms, including non standard ones.
- The objective is therefore a scalable pipeline combining:
 - symbolic regression to discover candidate laws,
 - symbolic algebra to test whether they remain analytically solvable.

Goals of the project

- 1 Fit NIHAO dark matter density profiles with **symbolic regression**, without imposing a fixed parametric family.
- 2 Assess the **analytic solvability** of the resulting expressions for derived quantities such as $M(< r)$, $\Phi(r)$, $\Sigma(R)$ and $\bar{\Sigma}(< R)$.
- 3 Move toward a **fully automated workflow** from raw simulation profiles to analytic validation and physical interpretation.

Two technical pillars

PhySO proposes analytic density laws from data; **Sage** certifies which physical quantities remain available in closed or semi-closed form.

Dark-matter halo profiles and their equations

$$M(< r) = 4\pi \int_0^r \rho(r') r'^2 dr' \quad (1)$$

$$\rho(r) = \frac{1}{4\pi r^2} \frac{dM(< r)}{dr} \quad (2)$$

Name	$\rho(r)$	Notes
NFW	$\frac{\rho_0}{(r/R_s)(1+r/R_s)^2}$	Cuspy, $(\gamma_{\text{in}}, \gamma_{\text{out}}) = (-1, -3)$
superNFW	$\frac{\rho_0}{(r/R_s)(1+r/R_s)^{5/2}}$	Steeper outskirts, sharper truncation
pISO	$\frac{\rho_0}{1+(r/R_s)^2}$	Cored, approximately isothermal
Burkert	$\frac{\rho_0 R_s^3}{(r+R_s)(r^2+R_s^2)}$	Cored, popular for dwarfs/LSBs
Lucky13	$\frac{\rho_0}{(1+r/R_s)^3}$	Cored analogue of NFW
Einasto	$\rho_0 \exp\left[-\frac{2}{\alpha}((r/R_s)^\alpha - 1)\right]$	Variable slope, standard in simulations
coreEinasto	$\rho_0 \exp\left[-\frac{2}{\alpha}((r/R_s + r_c/R_s)^\alpha - 1)\right]$	Einasto with explicit core radius
DiCintio	$\frac{\rho_0}{(r/R_s)^\alpha [1 + (r/R_s)^{1/\beta}]^{\beta(\gamma-\alpha)}}$	Feedback-calibrated double power law
gNFW	$\frac{\rho_0}{(r/R_s)^\gamma (1+r/R_s)^{3-\gamma}}$	Generalised inner slope γ
Dekel-Zhao	$\frac{\rho_0}{(r/R_s)^\alpha [1 + (r/R_s)^{1/2}]^{7-2\alpha}}$	Flexible two-slope family
pISO1	$\frac{1}{1+r^2}$	Fixed-parameter pseudo-isothermal
Exp	$\rho_0 \exp(-r/R_s)$	Single-scale exponential sphere
Exp1	$9.6 \exp(-r/1.4)$	Normalised exponential test case
Exp2	$\rho_0 \exp[-r/(R_{s,1} + R_{s,2})]$	Two-scale exponential sphere

Derived quantities

$$(1) \quad v_{\text{circ}}(r) = \sqrt{\frac{GM(< r)}{r}}$$

$$(2) \quad \Phi(r) = - \int_r^\infty \frac{GM(< r')}{r'^2} dr'$$

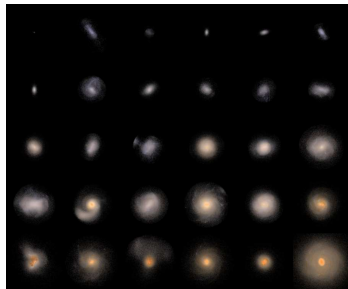
$$(3) \quad \sigma_r^2(r) = \frac{1}{\rho(r)} \int_r^\infty \rho(r') \frac{GM(< r')}{r'^2} dr'$$

$$(4) \quad \Sigma(R) = 2 \int_R^\infty \frac{\rho(r') r'}{\sqrt{r'^2 - R^2}} dr'$$

$$(5) \quad \bar{\Sigma}(< R) = \frac{2}{R^2} \int_0^R \Sigma(R') R' dR'$$

NIHAO: overview

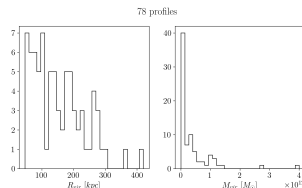
- Numerical Investigation of a **Hundred Astrophysical Objects**.
- Cosmological **zoom-in hydrodynamical** simulations sampling galaxies over a wide halo-mass range.
- Haloes selected from dark-matter-only boxes of 60, 20, and 15 h^{-1} Mpc.
- Each halo re-simulated at high resolution using the **zoom-in technique**.
- Flat Λ CDM cosmology: $H_0 = 67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3175$, $\sigma_8 = 0.8344$.
- Hydrodynamics with GASOLINE2: radiative cooling, star formation, stellar feedback (pre-SN + SN).



NIHAO: mass range, resolution & sample

Mass range & resolution

- Virial masses:
 $5 \times 10^9 - 2 \times 10^{12} M_{\odot}$ (dwarfs to Milky-Way scale).
- $\sim 10^6$ DM particles inside R_{vir} for every halo.



Distribution of the 78 NIHAO haloes in virial radius (left) and virial mass (right).

Sample used in this work

- 78 NIHAO haloes, $R_{\text{vir}} \sim 40\text{--}420$ kpc, $M_{\text{vir}} \sim 3.5 \times 10^9\text{--}2.8 \times 10^{12} M_{\odot}$.
- Spherically averaged 3-D DM density $\rho(r)$ at $z = 0$.
- Profiles normalised to dimensionless units before regression:

$$\frac{r}{R_{\text{vir}}}, \quad \frac{\rho}{\bar{\rho}_{\text{vir}}}, \quad \bar{\rho}_{\text{vir}} = \frac{3M_{\text{vir}}}{4\pi R_{\text{vir}}^3}.$$

From Sympy to Sage

- A first purely SymPy-based pipeline was sufficient for simple benchmarks but struggled on more complex expressions (slow execution, unevaluated integrals, ambiguous non-failures, ...)
- Sage was adopted as the main computer-algebra system because it wraps several mature symbolic backends under a Python-like interface.

Operational advantage

Sage can be scripted directly from Python, which makes it suitable for an automated loop with PhySO rather than a manual case-by-case study.

Structure of the Sage module and diagnostics

- Main task: given an analytic law for $\rho(r)$ or $M(< r)$, determine which derived quantities can be obtained in closed form.
- Derived quantities tested: $M(< r)$, $\rho(r)$, $v_{\text{circ}}(r)$, $\Phi(r)$, $\sigma_r^2(r)$, $\Sigma(R)$, and $\bar{\Sigma}(< R)$.
- Each computation is wrapped in a timeout so that one pathological expression cannot stall the full pipeline.
- A lightweight filter rejects results that still contain unevaluated integrals or undefined symbols.
- Successful symbolic expressions are exported to Python/NumPy/SciPy-readable strings for downstream use.

Analytic solvability of standard profiles

Key points

- The pipeline reproduces the expected hierarchy of tractability.
- Simple or steeply truncated profiles admit the richest set of closed-form derived quantities.
- Flexible families gain fitting freedom, but often lose analytic closure for Φ , σ_r^2 , Σ , or $\bar{\Sigma}$.
- Some partially tractable cases survive through standard special functions.

Profile	M	v_{circ}	σ_r^2	Φ	Σ	$\bar{\Sigma}$
NFW	✓	✓	×	✓	×	×
Super-NFW	✓	✓	✓	✓	×	×
ρ ISO	✓	✓	✓	×	✓	✓
Burkert	✓	✓	×	✓	×	×
Einasto	×	×	×	×	×	×
gNFW	✓	✓	×	×	×	×
Exponential	✓	✓	✓	✓	×	×

Representative summary adapted from the benchmark classification in the report.

Power-law based diagnostics

Motivation

Before running full Sage integration, quickly **flag physically pathological** candidate profiles (diverging mass, ill-defined potential, ...) at negligible cost.

Define the **asymptotic power-law index** at limit $a \in \overline{\mathbb{R}}^+$:

$$\sigma = \lim_{r \rightarrow a} \frac{-\log \rho(r)}{\log r}, \quad \text{so that} \quad \rho(r) \sim r^{-\sigma}.$$

- | | |
|-------------------------|------------------------|
| ① Finite at origin | ④ Velocity dispersion |
| ② Finite total mass | ⑤ Surface density |
| ③ Potential convergence | ⑥ Mean surface density |

Role in the pipeline

These diagnostics could act as a fast pre-filter before Sage: candidates failing basic convergence are discarded immediately.

Power-law based diagnostics

Figure 1: Summary of physical validity tests for parametric density profiles.

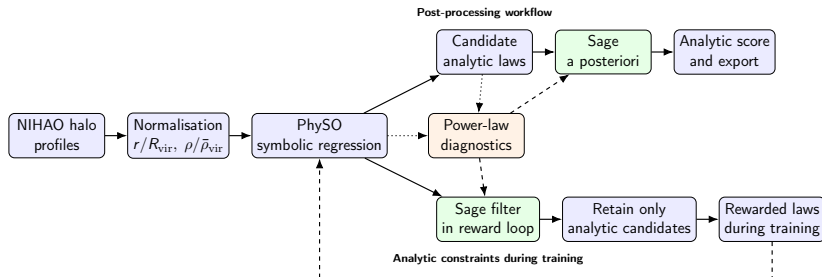
Profile	Finite at $r = 0$	Slope σ at ∞	$M(\infty) < \infty$	Φ finite	σ_r^2 finite	Σ finite	$\bar{\Sigma}$ finite
NFW	✓	3	✗ ($\sigma = 3$)	✓	✓	✓	✓
superNFW	✓	7/2	✓	✓	✓	✓	✓
pISO	✓	2	✗ ($\sigma = 2$)	✗ ($\sigma \notin (2, 3]$)	✓	✓	✓
pISO1	✓	2	✗ ($\sigma = 2$)	✗ ($\sigma \notin (2, 3]$)	✓	✓	✓
Burkert	✓	3	✗ ($\sigma = 3$)	✓	✓	✓	✓
Lucky13	✓	3	✗ ($\sigma = 3$)	✓	✓	✓	✓
Einasto	✓	∞	✓	✓	✓	✓	✓
coreEinasto	✓	∞	✓	✓	✓	✓	✓
DiCintio	$\sim (\alpha < 3)$	γ	$\sim (\gamma > 3)$	$\sim (2 < \gamma \leq 3)$	$\sim (\gamma > 1)$	$\sim (\gamma > 1)$	$\sim (\alpha < 3)$
gNFW	$\sim (\gamma < 3)$	3	✗ ($\sigma = 3$)	✓*	✓*	✓	$\sim (\gamma < 3)$
Dekel-Zhao	$\sim (\alpha < 3)$	7/2	✓	✓*	✓*	✓	$\sim (\alpha < 3)$
Exponential	✓	∞	✓	✓	✓	✓	✓
Exponential1	✓	∞	✓	✓	✓	✓	✓
Exponential2	✓	∞	✓	✓	✓	✓	✓

✓* Pass at large r , but finiteness at the origin undetermined (depends on the inner cusp exponent; see condition in column 1).
The case $\sigma = \infty$ corresponds to inverse super-polynomial growth at infinity.

The PhySO framework

- PhySO is a symbolic-regression engine designed to discover analytic relations directly from data while enforcing physical consistency.
- Instead of fitting a pre-chosen family, it searches in the space of mathematical expressions represented as trees.
- In this project we use the *class symbolic-regression* mode: one expression is fitted to the whole halo ensemble.
- Expression grammar includes arithmetic operators and elementary functions, with dimensional analysis used to suppress unphysical candidates.
- The reward is based on the fit error in $\log \rho$ – $\log r$ space, using dimensionless profiles and Poisson-weighted uncertainties.

The PhySO–Sage workflow



Symbolic regression fits to NIHAO

Setup: 8 runs, DMO & Hydro, $N_{\text{SPE}} \in \{1, 2, 3, 4\}$, $N_{\text{CLASS}} = 1$.

Convergence (Hydro, $N_{\text{SPE}} = 2$):

- Fast rise then plateau in R^2 .
- Population drifts toward shorter, physical expressions.
- Best-of-epoch profiles converge to a tight bundle.

Best hydro expression ($N_{\text{SPE}} = 2$):

$$\rho_{\text{SR}}(r) \propto \frac{r_s^4}{r^3(r_s - r)}$$

→ steep inner cusp ($\propto r^{-3}$) + outer break. $R^2 \simeq 0.53$.

Symbolic regression fits to NIHAO

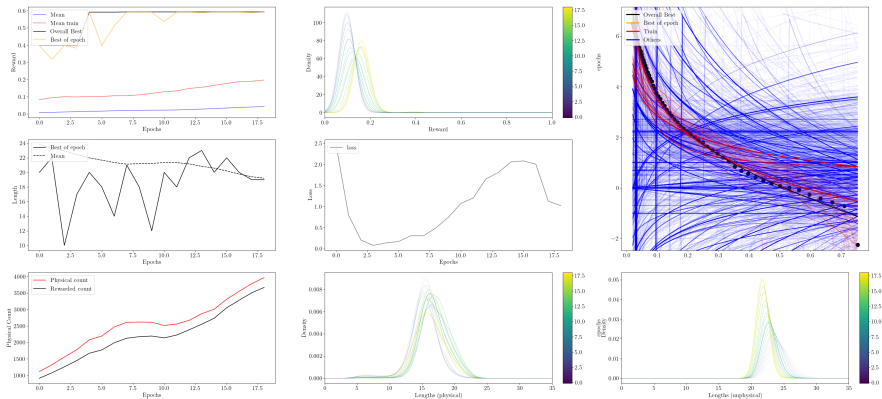


Figure 2: Convergence diagnostics for the Hydro run, $N_{\text{SPE}} = 2$.

Hydro vs DMO analytic complexity

- **Hydro**: tight band $R^2 \simeq 0.53$ – 0.59 .
- **DMO**: broader scatter in both R^2 and analytic score.
- Best-of-epoch for $N_{\text{SPE}} = 2$:
 - Hydro: $R^2 \simeq 0.53$ (median $\simeq 0.52$)
 - DMO: $R^2 \simeq 0.29$ (median $\simeq -0.30$)
- DMO best expression ($N_{\text{SPE}} = 2$):

$$\rho_{\text{dmo}} = \frac{\rho_0}{r} \left[\log \left(\frac{1}{1 - c_0 \frac{r}{r_s}} \right) \right]^{-1}$$

Take-away

Baryonic feedback does *not* obviously increase analytic complexity; hydro profiles are actually *easier* to fit with compact laws.

Hydro VS DMO analytic complexity

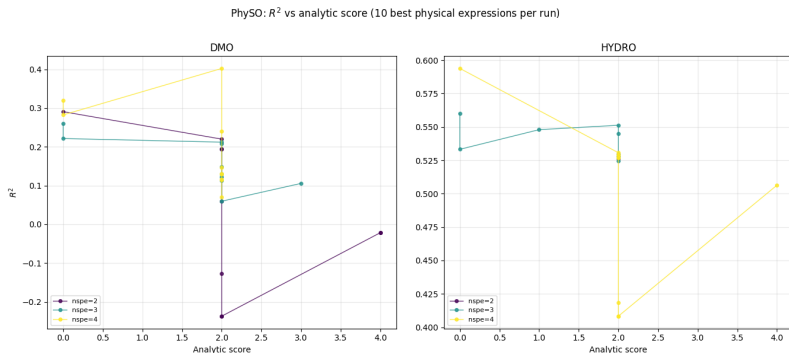


Figure 3: R^2 vs. analytic score for the 10 top physical expressions per run. Left: DMO. Right: Hydro. Colours = N_{SPE} .

Sage *a posteriori*

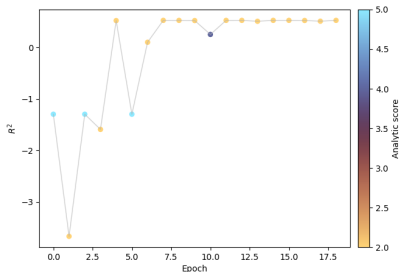


Figure 4: Best-of-epoch R^2 when Sage is applied only *a posteriori*.

- The symbolic-regression search is not constrained by analytic requirements during training.
- Candidate laws span a broad range of analytic scores, with high-score expressions appearing intermittently and mostly at late epochs.
- In the representative hydrodynamical run with $N_{\text{SPE}} = 2$, this configuration reaches the highest best-of-epoch R^2 values.

Sage in the reward loop

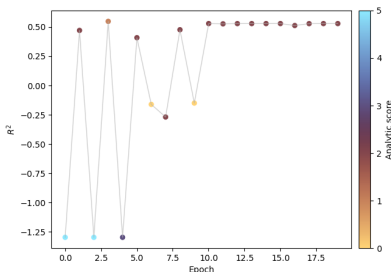


Figure 5: Same with analytic filtering by Sage enforced during training.

- Each candidate expression is first screened by Sage; only analytically admissible laws receive a non-zero reward.
- Converges onto families with nearly constant analytic score, showing that the mask is effective.
- Final fit quality remains slightly lower than in the *a posteriori* configuration.

Take-Away

Exploratory setup restricted that need refinement, but promising results opening application too many other works.

Discussion

- Symbolic regression recovers both standard halo profiles (NFW, Dekel–Zhao) and more original hybrid profiles.
- There is a trade-off between simplicity and accuracy: simple standard profiles are easier to interpret, while hybrid profiles fit simulations slightly better. The goal is therefore to find the best compromise between fit quality (R^2) and analytic tractability, using the PhySO + Sage framework.
- Some recovered expressions include logarithmic or exponential terms, creating intermediate asymptotic behaviours not captured by standard models. These hybrid forms provide a small but systematic improvement in fit quality compared to NFW/Einasto expressions.

Discussion

- Simple analytic profiles (NFW, Burkert, pISO) remain very useful because they provide closed-form expressions for key quantities like $M(< r)$, $\Phi(r)$ and sometimes $\Sigma(R)$.
- More accurate PhySO derived profiles often involve special functions or no closed form solutions, making them less practical for lensing and dynamical modelling.

Conclusion

- PhySO + Sage provides an automated framework to identify analytic dark matter halo profiles from NIHAO simulations.
- Symbolic regression finds both standard like and hybrid profiles, often improving the fit over NFW and Einasto models.
- Hydrodynamical haloes are not significantly more complex than DMO haloes in the explored configurations.
- More broadly, this work illustrates how machine learning and symbolic mathematics can be combined to discover physically meaningful laws in modern astrophysics.